BoneForge

Documentation

Version 8.5.0 · For VRChat Users

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Getting Started

What is BoneForge?

BoneForge is a Blender add-on that helps you prepare 3D avatars for VRChat, VRoid/VRM, and MMD. Think of it as a helper toolkit that sits inside Blender and handles the complicated parts of getting your avatar's skeleton set up correctly.

What Blender does: Blender is the 3D software where you edit your avatar's shape, textures, and movement system before uploading to VRChat. It is free and powerful, but can feel confusing at first.

What BoneForge adds: BoneForge adds panels and buttons to Blender that automate the most tedious steps — things like organizing bones, fixing names, setting up physics, and exporting in the right format.

New in 8.5.0: The open Blender build is version-aligned with the B4Artists release while remaining the standard Blender package. It exposes the complete CATS Material Atlas UV variation set for the CATS/Material Combiner/UVToolkit workflow, including Advanced Variation and Rotation Step, but it does not include B4Artists-exclusive rigging, control picker, lockout, or host-specific content.

8.5.0 also updates the export and validation workflow. VRChat, VRM, MMD, and Unreal exports now expose folder and file-name controls in the BoneForge panels instead of forcing you to rely only on Blender's file browser defaults. VRChat/Unity and Unreal FBX exports default to Embed Textures for easier material import, while VRChat export keeps helper/control-shape meshes out of the FBX unless Helper Meshes is enabled. The VRM panel now includes Lint Now plus Fix Humanoid Map, which can repair stale humanoid alias mapping before export without renaming your bones.

New in 8.3.1 (Auto-Rig IK and control density update): The Auto-Rig body generator now creates dedicated, non-deforming IK target bones named **`hand_ik.L`**, **`hand_ik.R`**, **`foot_ik.L`**, and **`foot_ik.R`** when IK is enabled. This gives hands and feet proper endpoint controls instead of using deform bones as IK targets. The wizard also exposes controller-density options for Spine Segments and Neck Segments, so generated rigs can use smoother torso and neck chains when needed.

New in 8.2.1: BoneForge continues the 8.x Auto-Rig and avatar-prep line with the reorganized 7.2.1 source layout and later rig-generation improvements. Existing 7.x documentation remains broadly applicable, but the Auto-Rig generation controls documented here reflect the 8.3.1 build.

New in 7.1.3 (preference label cleanup): Two add-on preference toggles were renamed so they now match the sidebar tab they control. "VRChat Avatar Tools" is now labelled CATS (matches the CATS sidebar tab). "Task Board & Sidebar" is now labelled Rig Builder (matches the Rig Builder sidebar tab). No tools were removed — only the on/off labels in Edit > Preferences > Add-ons > BoneForge changed.

New in 7.1.1: BoneForge now includes the CATS Plugin — a complete suite of model preparation tools specifically designed for getting VRChat avatars clean, optimized, and fully configured. CATS lives in its own sidebar tab and uses a pipeline system that guides you through the right order of operations every time.

What BoneForge cannot do:

- It cannot model or sculpt your avatar's body shape
- It cannot create textures or materials from scratch
- It cannot upload to VRChat directly (you still need the VRChat Creator Companion / SDK)

Open Blender vs B4Artists Build Split

BoneForge has two 8.5.0 builds with different package boundaries. Matching version numbers do not mean the same payload.

Host	Standard Blender	Bforartists only
Add-on identity	**`BoneForge`**	**`BoneForge BFA`**

Repository	**`Axleonex/BoneForge_ALTERNATIVE_CATS_for_5.0_Blender`**	**`Axleonex/BoneForge_B4Artists`**
Release zip	**`BoneForge-8.5.0.zip`**	**`BoneForge-BFA-8.5.0.zip`**
CATS avatar cleanup	Included	Included
Material Combiner	Included	Included
UVToolkit-derived Material Combiner controls	Included, including Advanced Variation and Rotation Step	Included, same CATS / Material Combiner / UVToolkit behavior
Basic BoneForge and Mixamo-style avatar helpers	Included	Included
BFA host lockout	Not included	Included
Production control-rig construction	Not included	B4Artists-exclusive
Smart landmark / joint detection suite	Not included as a BFA production suite	B4Artists-exclusive
Animator control layer	Not included	B4Artists-exclusive
Control Picker / rig UI	Not included	B4Artists-exclusive
Advanced retargeting core and source maps	Not included as a BFA production suite	B4Artists-exclusive
Profile-driven game export	Not included as a BFA production suite	B4Artists-exclusive
BFA marker files	Must not contain **`bfa_guard.py`** or **`BFA_EXCLUSIVE.md`**	Must contain **`bfa_guard.py`** and **`BFA_EXCLUSIVE.md`**

The open Blender build is the non-exclusive standard-Blender package. It now receives the complete CATS, Material Combiner, and UVToolkit-derived workflow. The B4Artists build remains the exclusive package for the production rigging suite, host lockout, control rig builder, animator controls, control picker, advanced retarget/export systems, and Bforartists-specific packaging.

Installing BoneForge

What you need before starting:

- Blender 4.0 or newer (download free at blender.org)

- The BoneForge `.zip` file

Steps:

1. Open Blender
2. Go to Edit > Preferences (top menu bar)
3. Click Add-ons on the left side
4. Click Install from Disk (top right of the Add-ons panel)
5. Navigate to your BoneForge `.zip` file and select it
6. Click Install Add-on
7. Find "BoneForge" in the add-on list and check the checkbox to enable it
8. Click the arrow next to BoneForge to expand its settings — you can choose which tools to enable

You should see: A new panel called "BoneForge" appear in the right sidebar of the 3D viewport (press N to open/close the sidebar). You will also see a separate CATS tab in the same sidebar.

Finding BoneForge in Blender

When BoneForge is installed, here is where everything lives:

The Sidebar (most common): Press N in the 3D viewport to open a panel on the right side. You will see tabs including BoneForge's tools organized by task.

Tabs you will use most:

- Rig Builder — Build a new rig from scratch
- Setup Rigging — Retargeting and Rigify tools
- Skin — Weight and deformation tools
- VRChat — Everything for VRChat export
- Review / Animate — Bone visibility, pose library, validation
- CATS — Model cleanup, visemes, eye tracking, and full VRChat prep pipeline (new in 7.1.1)

The CATS tab is a separate tab from the main BoneForge tabs. Scroll down the sidebar tab list if you do not see it immediately — it appears after the BoneForge tabs.

Fix Model First, Every Time: When using the CATS tab, always start with Fix Model before running any other CATS tool. The CATS pipeline uses a Ledger to track which steps you have completed. Each tool checks the Ledger and will warn you if you try to run it out of order. See CATS Plugin — Before You Begin: The Pipeline Order for the full explanation.

The N-Panel Hotkey: You can press Ctrl+Shift+R in the 3D viewport to pop up BoneForge's quick panel wherever your cursor is, without navigating to the sidebar.

Where Should I Start?

Choose the description that best fits you:

> "You are a person who has a brand new 3D model file (FBX, OBJ, or Blender file) and wants to rig it for VRChat from scratch." Go to Guide 1: Get Your First Avatar Into VRChat

> "You are a person who made your avatar in VRoid Studio and exported a VRM file." Go to Guide 2: Bring a VRoid Avatar into VRChat

> "You are a person who has an MMD model (PMX file) and wants to use it in VRChat." Go to Guide 3: Bring an MMD Avatar into VRChat

> "You are a person who already has a rigged avatar but it won't upload because the bones have wrong names." Go to Guide 4: Fix Your Avatar's Bone Names

> "You are a person who has a rigged avatar and wants to get it fully VRChat-ready — lip sync, eye tracking, clean mesh, all at once." Go to Guide 13: Get Your Avatar VRChat-Ready with CATS

> "You are a person who wants to fix a specific problem." Go to the Quick-Fix Index

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Quick Guides

Guide 1: Get Your First Avatar Into VRChat

> Time: About 45–60 minutes for a complete first run Result: A fully rigged, VRChat-ready avatar exported as an FBX file

Before you begin — check these:

- [ ] Your 3D model is imported into Blender (File > Import, choose your format)
- [ ] The model is in a T-pose (arms out to the sides, body upright) — or close to it
- [ ] The model has no obvious broken geometry (no random floating triangles)
- [ ] You can see your model in the 3D viewport as a solid grey shape

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Step 1 — Open the Rig Builder

In the right sidebar (press N if it is not visible), click the Rig Builder tab. You will see three options: Quick Rig, Wizard, and Mannequin.

Click Wizard to start the guided rigging process.

What you should be seeing: A panel that says "Start" with a button to begin the wizard.

Step 2 — Start the Wizard and Select Your Mesh

Click Start Wizard. The wizard will ask you to select your mesh (your avatar's 3D body shape).

Click on your avatar in the 3D viewport to select it, then click Confirm Selection in the wizard panel.

What you should be seeing: The wizard advances to a screen showing "Rig Type."

> Rig type = the style of skeleton BoneForge will create. For VRChat avatars that look like humans, choose Human.

On the review/generation screen, BoneForge also shows Generation Options:

- Kinematics — Choose IK + FK, IK Only, or FK Only. Use IK + FK for most VRChat avatars because it gives you both endpoint controls and traditional rotation controls.
- Generate Control Shapes — Creates easier-to-select viewport controls for pose bones.
- Spine Segments — Controls how many bones are generated in the spine chain. Higher values allow smoother torso bending.
- Neck Segments — Controls how many bones are generated in the neck chain. Higher values are useful for long necks, stylized avatars, or creatures.

When IK is enabled, generated rigs include non-deforming target bones named **`hand_ik.L`**, **`hand_ik.R`**, **`foot_ik.L`**, and **`foot_ik.R`**. These belong in the IK controls collection and should not be weight-painted to the mesh.

Step 3 — Set Finger Count

The wizard asks how many fingers your avatar has on each hand. For a standard human avatar, this is 5 per hand. If your avatar has fewer or stylized hands, adjust accordingly.

Step 4 — Place Body Markers

This is the most important step. You are placing dot markers on your avatar to show BoneForge where each major body part is located. Think of it like pinning a map — you are telling BoneForge "the pelvis is here, the head is here," and BoneForge figures out all the bone positions from those pins.

How to place a marker:

1. Select the marker name from the list (e.g., "Pelvis")
2. Click Place Marker
3. Click on the correct spot on your avatar in the 3D viewport
4. The marker dot turns green when confirmed

Tip — Use Auto-Detect: Click Guess Body Markers and BoneForge will attempt to place all markers automatically based on your mesh shape. Check that each marker is green and in a reasonable position. You can click any marker and use Move Marker to adjust.

Tip — Use Symmetry: Enable Mirror to automatically place the right-side marker whenever you place a left-side marker. This saves time for arms, legs, shoulders, and feet.

Required body markers (7 total): Head Top, Neck Base, Pelvis, Left Wrist, Right Wrist, Left Ankle, and Right Ankle.

Optional precision markers: Shoulders, elbows, hips, knees, toes, and heels can be placed manually for more accurate limb proportions. If you skip them, BoneForge derives those joint positions from the required markers.

What you should be seeing: All required body markers showing green dots on your avatar. Optional markers can also be green if you placed them manually.

Step 5 — Place Face Markers (Optional)

If your avatar has a face that you want to animate (blinking, expressions, lip sync), place the face markers too. These are optional but strongly recommended for VRChat.

Click Guess Face Markers for automatic placement, then adjust as needed.

Step 6 — Place Finger Markers

Click Guess Finger Markers for automatic finger placement. BoneForge will trace each finger chain from knuckle to fingertip.

Step 7 — Review and Generate

Click Next to reach the Review screen. BoneForge shows a summary of what it is about to create. Click Generate Rig.

What you should be seeing: BoneForge creates a skeleton (shown as orange bones) inside your avatar. Your avatar's skin should be automatically attached to the deform bones so it deforms when you move the rig. If you generated IK controls, you should also see separate hand and foot IK targets; these are controls, not skinning bones.

> Skin/weight painting = the process of deciding which parts of your avatar's mesh follow which bones. BoneForge handles this automatically in the initial pass, but you can refine it later using the Weight Tools (see Feature Reference).

Step 8 — Fix Bone Names for VRChat

After generating, your bones need to follow VRChat's naming rules. Go to the VRChat tab in the sidebar and click Fix Bone Names > Auto-Detect and Rename.

What you should be seeing: All bones in the list show green checkmarks.

Step 9 — Map to VRChat Humanoid

Still in the VRChat tab, find the Humanoid Mapper section. Click Auto-Map Humanoid. This connects each of your avatar's bones to VRChat's humanoid system (the system VRChat uses to make avatars move in sync with your real-world movements).

Run Validate Humanoid to check for any remaining issues.

What you should be seeing: A list of humanoid slots (Hips, Spine, Head, etc.) each showing a bone name next to it.

Step 10 — Export for VRChat

Go to the VRChat tab Export section. Click Export to VRChat (FBX).

Choose a save location and click Export.

What you should be seeing: A `.fbx` file saved to your chosen location. This file is what you import into the VRChat Creator Companion.

What This Unlocks: You now have a fully rigged VRChat avatar file. From here you can add hair physics (Guide 6), attach clothing (Guide 7), set up lip sync (Guide 8), and optimize performance (Guide 9). For a one-stop cleanup and VRChat configuration workflow using the new CATS tools, see Guide 13.

Guide 2: Bring a VRoid Avatar into VRChat

> Time: About 15–25 minutes Result: Your VRoid `.vrm` file ready for VRChat export

Before you begin — check these:

- [ ] You have exported your avatar from VRoid Studio as a `.vrm` file
- [ ] BoneForge is installed and enabled
- [ ] The VRM bridge add-on is installed (see below)

Step 1 — Install the VRM Bridge

BoneForge needs a helper add-on to open VRM files. In the BoneForge sidebar, go to the VRM section and click Install VRM Add-on Automatically. If that fails, click Open VRM Website to download the official VRM add-on manually and install it the same way you installed BoneForge.

Step 2 — Import Your VRM File

Go to File > Import > VRM (.vrm) and select your VRoid file.

What you should be seeing: Your VRoid character appears in Blender with its skeleton already in place.

Step 3 — Auto-Map to VRChat Humanoid

Go to the VRChat tab in the BoneForge sidebar. Click Auto-Map Humanoid. VRoid avatars follow a standard skeleton format, so this usually completes automatically without any manual adjustments.

Step 4 — Fix Bone Names

Click Fix Bone Names > Auto-Detect and Rename. VRoid uses its own naming system; this converts the names to what VRChat expects.

Step 5 — Set Up Visemes (Lip Sync)

VRoid avatars already have blend shapes (the face movements for expressions and lip sync) built in. Go to the VRChat tab Visemes and click Auto-Map Visemes. BoneForge will match VRoid's shape keys to VRChat's 15 lip-sync phonemes automatically.

Step 6 — Export

Click Export to VRChat (FBX) in the VRChat Export section.

What This Unlocks: Your VRoid avatar is now ready to import into the VRChat Creator Companion. You can also add hair physics (Guide 6) and optimize performance (Guide 9) before uploading.

Guide 3: Bring an MMD Avatar into VRChat

Time: About 20–30 minutes Result: Your MMD `.pmx` model ready for VRChat

Before you begin — check these:

- [ ] You have a `.pmx` or `.pmd` MMD model file
- [ ] BoneForge is installed
- [ ] The MMD Tools add-on is installed (see Step 1)

Step 1 — Install MMD Tools

In the BoneForge sidebar, scroll to the MMD section and click Install MMD Tools Automatically. If that fails, click Open MMD Website to download MMD Tools manually.

Step 2 — Import Your PMX File

Go to File > Import > MikuMikuDance Model (.pmx/.pmd) and select your model.

What you should be seeing: Your MMD character appears in Blender with Japanese-style bone names.

Step 3 — Fix Bone Names

MMD uses Japanese bone names that VRChat cannot understand. In the VRChat > Naming section, click Detect Convention. BoneForge will recognize the MMD naming style. Then click Translate Bone Names to convert them to VRChat-compatible names.

Alternatively, use the CATS tab Bone Name Translation tool, which supports automatic language detection for Japanese, Chinese, Korean, Portuguese, Spanish, and French bone names in a single click.

Step 4 — Clean Up the Model

MMD models often have extra geometry and duplicate vertices. Go to VRChat > Cleanup and click:

- Fix Model — removes problem geometry
- Join Meshes — combines body parts into one mesh (recommended for VRChat)

- Remove Unused Vertex Groups — removes empty bone assignments

For a guided, pipeline-ordered version of this cleanup, use the CATS tab and follow the pipeline order described in Guide 13.

Step 5 — Map to VRChat Humanoid

Click Auto-Map Humanoid in the VRChat Humanoid section. MMD's skeleton is similar to VRChat's, so most slots fill automatically. Fix any unmatched slots manually by clicking the slot and choosing the correct bone from the dropdown.

Step 6 — Export

Click Export to VRChat (FBX).

What This Unlocks: Your MMD avatar is now ready for VRChat. You can add hair physics (Guide 6) and set up lip sync (Guide 8) before uploading.

Guide 4: Fix Your Avatar's Bone Names

> Time: About 5–15 minutes Result: All bones renamed to VRChat-compatible names

Before you begin — check these:

- [ ] Your avatar is open in Blender with its skeleton (armature) visible
- [ ] You know roughly what naming format your avatar uses (e.g., Mixamo, VRoid, Unity, custom)

Step 1 — Detect Current Naming Style

Go to the VRChat tab Naming section. Click Detect Convention. BoneForge will analyze your bone names and show you what style it detected (Mixamo, Ready Player Me, Unity, or Custom).

For models with bone names in Japanese, Chinese, Korean, Portuguese, Spanish, or French, use the CATS tab Bone Name Translation tool instead. It auto-detects the source language and converts everything to English VRChat names in one step.

Step 2 — Auto-Translate (Recommended)

If BoneForge detected a known naming style, click Translate Bone Names. This renames everything automatically.

What you should be seeing: The bone list shows VRChat-compatible names like **`Hips`**, **`Spine`**, **`Chest`**, **`LeftUpperArm`**, etc.

Step 3 — Manual Fixes (If Needed)

If some bones were not automatically renamed, use the tools in the Batch Rename section:

- Find and Replace — Type the old text in the left box, the new text in the right box, click Apply
- Add Prefix — Adds text to the start of all bone names (e.g., turning **`Arm`** into **`Left_Arm`**)
- Add Suffix — Adds text to the end of all bone names
- Remove Prefix / Remove Suffix — Strips added text

Step 4 — Save as a Preset (Optional)

If you have a custom avatar with unique naming, click Save Custom Preset after getting everything named correctly. This saves your naming rules so you can apply them instantly to future avatars.

What This Unlocks: Correct bone names allow the Humanoid Mapper (Guide 5) to work automatically. Without correct names, VRChat's avatar system cannot recognize how your character should move.

Guide 5: Map Your Avatar to VRChat's Body System

> Time: About 10–15 minutes Result: Your avatar's skeleton connected to VRChat's humanoid movement system

Before you begin — check these:

- [ ] Your avatar's bones are named correctly (see Guide 4 if not)
- [ ] Your avatar is in Blender with its skeleton selected
- [ ] You are in Object Mode (check the dropdown in the top-left of the viewport)

Step 1 — Open the Humanoid Mapper

Go to the VRChat tab Humanoid section.

Step 2 — Auto-Map

Click Auto-Map Humanoid. BoneForge scans your skeleton and fills in the required body slots automatically.

> A humanoid slot is like a labeled hook that VRChat uses: "Hips goes here, Head goes here, Left Hand goes here." BoneForge matches your bones to these hooks.

What you should be seeing: A list of slots (Hips, Spine, Chest, Neck, Head, LeftUpperArm, etc.) each with a bone name filled in next to it.

Step 3 — Check for Errors

Click Validate Humanoid. BoneForge checks all required slots are filled and the hierarchy makes sense.

- Green = correct
- Yellow = warning (not required but recommended)
- Red = error (must fix before export)

Step 4 — Fix Errors Manually

If any red errors appear, click the error message. BoneForge will highlight the problem bone. Use the dropdown next to the slot to select the correct bone manually.

Step 5 — Set Up Eye Tracking (Optional)

If your avatar has eye bones, go to the Eye Setup section. Click Fix Eye Bones to ensure both eye bones are named and positioned correctly. Click Auto-Map Blink Shapes to connect your blink animations.

For a more complete eye tracking setup including constraint creation and VRChat LeftEye/RightEye bone naming, use the CATS tab Eye Tracking Setup tool described in Guide 13, Phase 3.

What This Unlocks: Once humanoid mapping is complete, your avatar will move properly in VRChat — IK (inverse kinematics, the system that makes your in-game hands follow your real controllers) will work, and your avatar will track your real-world movements correctly.

Guide 6: Add Hair Physics

> Time: About 15–25 minutes Result: Your avatar's hair (and any soft accessories) bouncing and swaying naturally in VRChat

Before you begin — check these:

- [ ] Your avatar has hair bones already in the skeleton (bones that form chains from the scalp outward)
- [ ] Your avatar is open in Blender with the skeleton selected

> PhysBone = VRChat's name for a component that makes bones bounce and swing as if they have weight. BoneForge creates these automatically from your hair bone chains.

Step 1 — Detect Hair Chains

Go to the VRChat tab Hair Physics section. Click Detect Hair Chains.

BoneForge scans your skeleton for chains of bones that look like hair (multiple bones chained end-to-end, branching from a root). It lists all the chains it found.

What you should be seeing: A list of chain names, each with a start bone and a number of chain links.

Step 2 — Review Detected Chains

Look through the list. If BoneForge detected something that is not hair (like a tail bone you want to handle separately, or a belt chain), you can remove it from the list by clicking the minus button next to it.

Step 3 — Choose a Physics Preset

Select a preset that matches how you want your hair to feel:

- Stiff — Hair that barely moves, like a helmet or rigid braid
- Normal — Natural flowing hair with moderate bounce
- Bouncy — Very loose, floaty hair with lots of movement

Step 4 — Generate PhysBone Components

Click Generate Hair PhysBones. BoneForge creates the physics components on all detected chains using your chosen preset.

What you should be seeing: Each hair chain in the list now shows a physics icon.

Step 5 — Fine-Tune Physics (Optional)

Click on any chain in the list and use the sliders to adjust:

- Stiffness — How much the bone resists bending (higher = stiffer)
- Damping — How quickly the swinging slows down (higher = less bouncy, more floaty)
- Gravity — How much the hair pulls downward (negative values pull upward)
- Drag — Air resistance (higher = slower, smoother movement)
- Collision Radius — How thick the physics collision zone around each bone is

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Step 6 — Add Colliders (Recommended)

Colliders are invisible shapes that prevent hair from clipping through your avatar's head and body. Click Place Default Colliders to automatically add standard collision shapes to your head, shoulders, and chest.

What you should be seeing: Small sphere or capsule shapes appearing around your avatar's head and upper body.

Step 7 — Preview (Optional)

Click Play Physics Preview to simulate the hair movement in Blender's viewport. Click Stop when done.

What This Unlocks: Your avatar's hair will now move naturally in VRChat when you turn your head, jump, or dance. You can apply the same process to tails, dangling accessories, or any other chain of bones you want to swing freely.

Guide 7: Attach Clothing That Moves With Your Body

> Time: About 20–30 minutes Result: Clothing mesh fully attached to your avatar's skeleton, moving correctly with your body

Before you begin — check these:

- [ ] Your base avatar is open in Blender and has a completed skeleton
- [ ] Your clothing item is imported into the same Blender file as a separate mesh object
- [ ] The clothing roughly fits around your avatar's body (it does not need to be perfect)

Step 1 — Open Clothing Tools

Go to the VRChat tab Clothing section.

Step 2 — Add Your Clothing to the List

Click Add Clothing Item and select your clothing mesh from the dropdown. Repeat for each clothing piece you want to attach.

Step 3 — Match Bones

Click Match Bones with your clothing item selected. BoneForge compares your clothing's skeleton (if it has one) to your base avatar's skeleton and creates a mapping between them.

If your clothing came with its own skeleton, BoneForge tries to find the equivalent bone in your base skeleton. For example, a "left arm" bone in the clothing skeleton gets matched to your avatar's left arm bone.

What you should be seeing: A list of bone pairs showing clothing bone avatar bone.

Step 4 — Review and Fix Mismatches

Any unmatched bones appear in yellow. For each unmatched bone, click the dropdown next to it and manually select the closest bone from your base skeleton.

For clothing without its own skeleton, skip this step.

Step 5 — Merge Clothing

Click Merge Clothing. BoneForge transfers the clothing mesh's weights (the assignments that decide which bones move which part of the mesh) to your base skeleton.

> Weights (also called weight painting) = numbers that tell each vertex (point) of your mesh how much it should be moved by each bone. If your left sleeve has weight on the left arm bone, moving the left arm bone will pull the sleeve with it.

What you should be seeing: Your clothing is now listed under your main avatar skeleton in the scene. Moving a skeleton bone should move both the body and the clothing together.

Step 6 — Check for Clipping

Use the Detect Collisions button to scan for areas where the clothing mesh is poking through the body mesh. Adjust colliders or refine weights in problem areas.

What This Unlocks: Your avatar now has clothing that moves naturally with your body. You can repeat this process for each outfit piece. For advanced weight adjustments, see the Weight Tools section in the Feature Reference.

Guide 8: Set Up Lip Sync

> Time: About 10–20 minutes Result: Your avatar's mouth moving correctly when you speak in VRChat

Before you begin — check these:

- [ ] Your avatar's head mesh has mouth shape keys (also called blend shapes or morph targets — these are the different mouth positions for speaking)
- [ ] You know roughly what the shape keys are named (check the Shape Keys panel in Blender's Properties panel on the right side)

> Viseme = a specific mouth shape that corresponds to a sound. "AA" is for the "ahh" sound, "OH" is for the "ohhh" sound, etc. VRChat needs 15 specific visemes to drive your avatar's lip sync. Shape key / blend shape = a saved version of your mesh in a different position. Your mouth-open shape key is a saved version of your mesh with the mouth open.

Step 1 — Open the Viseme Mapper

Go to the VRChat tab Visemes section.

Step 2 — Auto-Map Visemes

Click Auto-Map Visemes. BoneForge scans your mesh's shape keys and tries to match them to VRChat's 15 phoneme slots by name.

What you should be seeing: Most of the 15 phoneme slots filled with shape key names.

Step 3 — Fill in Missing Slots

For any empty phoneme slots, click the dropdown next to the slot and select the closest matching shape key from your mesh. Common matches:

`aa`	"ahh"	mouth_open, A, aa
`oh`	"ohh"	mouth_o, OH, oh
`ch`	"ch" / "sh"	mouth_ch, CH
`mm`	lips together (M, B, P)	mouth_m, MM, lips_together

ss	"sss" / "zzz"	mouth_s, SS

Alternative — CATS Viseme Generator: If your avatar does not have existing viseme shape keys, use the CATS tab Viseme Generator to create all 15 VRChat visemes from scratch using just three base shapes (A, O, CH). See Guide 13, Phase 2 for a step-by-step walkthrough.

Step 4 — Preview a Viseme

Click any phoneme name to preview what that mouth shape looks like on your avatar. Click it again to return to neutral.

Step 5 — Set Up Face Tracking (Optional)

If you want VRChat's face tracking to work with your avatar, enable Face Tracking in the Face Tracking section and adjust the expression smoothing slider.

What This Unlocks: Your avatar's mouth will now move when you talk in VRChat, making you look far more natural in conversations. Expression and emotion systems build on top of the shape keys you just mapped.

Guide 9: Improve Your Avatar's Performance

> Time: About 15–25 minutes Result: Avatar with a better VRChat performance rating and faster loading time for other players

Before you begin — check these:

- [ ] Your avatar is complete (rigged, clothed, lip sync set up)
- [ ] You know your target performance tier (Good or Excellent for most users)

> Performance rank = VRChat's rating system for how demanding your avatar is. Very Poor avatars may be hidden by other players. Good or Excellent avatars load fast and are always visible.

Step 1 — Check Current Performance Rank

Go to the VRChat tab Performance section. Click Calculate Rank. BoneForge shows your current estimated rank and the specific numbers causing it (polygon count, material count, bone count, etc.).

Step 2 — Clean Up the Mesh

In the Cleanup section, run these in order:

1. Fix Model — Removes duplicate geometry and fixes common mesh errors
2. Remove Unused Shape Keys — Deletes blend shapes that are not mapped to anything (frees memory)
3. Remove Unused Vertex Groups — Removes empty bone weight assignments
4. Remove Zero-Weight Bones — Deletes bones that do not move any part of the mesh

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Step 3 — Reduce Polygon Count (If Needed)

If your polygon count is too high, use the Decimation tool:

1. Move the Decimation Ratio slider (0.5 = halve the polygon count, 0.8 = remove 20%)
2. Click Preview Decimation to see the result without committing
3. Click Apply Decimation when you are happy with the result

> Start at 0.8 and work downward — small reductions rarely affect visible quality but can significantly improve performance.

Step 4 — Merge Materials (Optional)

If your avatar uses many different materials (color zones, texture sheets), use the Material Atlas section to combine them:

1. Click Analyze to see your current material layout
2. Click Add Group to create groups of materials to merge
3. Choose your atlas resolution (2048 recommended for most avatars)
4. Click Bake Atlas — BoneForge combines the materials into one texture sheet
5. Click Accept to apply, or Revert if you are not happy with the result

The CATS tab's Material Atlas Combiner offers the same Accept/Revert workflow in a streamlined interface — see CATS Tool Reference: Material Atlas Combiner.

Step 5 — Recalculate Rank

Click Calculate Rank again to see your improved performance score.

What This Unlocks: A better performance rank means more players will see your avatar without it being blocked. An Excellent or Good avatar loads quickly, consumes less GPU, and is always visible to other players by default.

Guide 10: Save and Reuse Poses

Time: About 5–10 minutes Result: A saved library of poses you can apply to your avatar with one click

Before you begin — check these:

- [ ] Your avatar has a completed skeleton in Blender
- [ ] You are in Pose Mode (click your armature, then press Ctrl+Tab and choose Pose Mode from the menu, or use the dropdown in the top-left of the viewport)

Step 1 — Open the Pose Library

Go to the Review tab in the BoneForge sidebar and find the Pose Library section.

Step 2 — Pose Your Avatar

Move your avatar's bones into the position you want to save. Rotate arms, tilt the head — any combination of bone positions becomes a pose.

Step 3 — Save the Pose

Click Save Pose. A dialog appears asking for a name and category. Type something descriptive (e.g., "Peace Sign" or "Thinking") and an optional category (e.g., "Greetings", "Action").

Click OK. A thumbnail of the current viewport is captured automatically.

Step 4 — Apply a Saved Pose

Click the thumbnail image of any saved pose in the Pose Library panel. Click Apply Pose to snap your avatar to that position.

- Apply Blended — Applies the pose at partial strength (a slider from 0% to 100%), great for mixing two poses together
- Apply Mirrored — Applies the pose flipped left-to-right, giving you a matching pose for the other side

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Step 5 — Export and Import Poses

Click Export Poses to save your pose library to a `.bfpose` file — a small file you can keep with your project or share with others. Click Import Poses to load poses from a `.bfpose` file.

What This Unlocks: A personal pose library you can use across projects. You can build a full set of reference poses, blend between them for animation keyframing, or share poses with other BoneForge users.

Guide 11: Merge Two Rigs Together

> Time: About 25–40 minutes depending on complexity Result: Two separate skeletons combined into one, with all weights preserved

Before you begin — check these:

- [ ] Both the base avatar and the secondary character (clothing rig, accessory rig, etc.) are in the same Blender file
- [ ] Both have been rigged and weighted

> Rig merge = the process of absorbing one skeleton into another so you end up with a single combined skeleton. Useful when you have a body rig and a hair/outfit rig that need to become one.

Step 1 — Open Bone Merge

Go to the Review tab Bone Merge section. Or find it in the sidebar's Bone Merge tab.

Step 2 — Stage 1: Analyze

Select your Target Armature (the main skeleton that will survive) and your Source Armature (the secondary skeleton being absorbed).

Click Analyze. BoneForge compares the two skeletons and creates a diff table showing:

- Matched — bones that exist in both and align correctly
- + Source Only — bones from the secondary skeleton with no match in the main skeleton
- − Target Only — bones in the main skeleton not found in the secondary

Click Acknowledge after reviewing. This unlocks Stage 2.

Step 3 — Stage 2: Resolve Names

For each Source Only bone (marked with +), you need to decide what to do with it:

- Rename it to match an existing bone in the main skeleton if they serve the same purpose
- Mark as Unique if it is a new bone that has no equivalent (it will be added to the main skeleton as-is)

Click Normalize to automatically rename all the standard bones (spine, arms, legs, etc.) that BoneForge recognizes.

For bones BoneForge cannot recognize, click Propose to get a suggested name based on the bone's position, then adjust manually.

Click Verify when all Source Only bones are resolved. This unlocks Stage 3.

Step 4 — Stage 3: Merge

Click Dry Run first. This shows you a preview of what the merge will do without making any changes. Review the report.

When satisfied, click Execute Merge. BoneForge automatically creates a backup of both armatures before merging, then combines them.

What you should be seeing: One armature in your scene containing all bones from both skeletons, with all meshes properly weighted.

What This Unlocks: A single unified rig that is easier to export, edit, and work with. Required for any avatar that has separate clothing or accessory rigs.

Guide 12: Fix Upload Problems

> Time: About 5–15 minutes depending on the issue

Before you begin: Identify your specific problem from the list below and jump to that section.

"Upload failed — bones not recognized"

Your bones have names VRChat does not understand. Go to Guide 4: Fix Your Avatar's Bone Names

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"Avatar is in T-pose / not moving with me in VRChat"

The humanoid mapping is missing or incorrect. Go to Guide 5: Map Your Avatar to VRChat's Body System

"Mesh is deforming weirdly / skin stretching wrong"

The bone weights need adjustment. See Weight Transfer and Weight Mirror in the Feature Reference.

"Hair is clipping through the head"

Hair colliders are missing or too small. Go to Guide 6: Add Hair Physics, Step 6.

"Avatar is Very Poor performance and being blocked by other players"

Go to Guide 9: Improve Your Avatar's Performance

"Lip sync not working"

Visemes are not mapped correctly. Go to Guide 8: Set Up Lip Sync

"Rig validator shows red errors"

Go to the Review tab Rig Validator section. Click Run Validation. For each red error, click the error message — BoneForge selects the problem bone for you. Read the error description and follow its suggestion, or check the Quick-Fix Index.

"VRM import not working"

The VRM bridge add-on may not be installed. Go to Guide 2, Step 1.

"MMD import not working"

The MMD Tools add-on may not be installed. Go to Guide 3, Step 1.

Guide 13: Get Your Avatar VRChat-Ready with CATS

> Time: About 20–35 minutes for a full pipeline run Result: A clean, optimized avatar with lip sync, eye tracking, and correct transforms — ready for VRChat upload

Before you begin — read this first:

The CATS tools use a Pipeline system. Each phase must be completed in the correct order. The CATS sidebar shows a Ledger — a row of checkmarks that tracks which phases you have completed. A grayed-out tool means its required earlier step has not been checked off yet.

Always start with Fix Model. Every time. Without exception.

Read CATS Plugin — Before You Begin: The Pipeline Order before continuing if you have not already.

Before you begin — check these:

- [ ] Your avatar is imported into Blender and visible in the 3D viewport
- [ ] The CATS tab is visible in the N-panel sidebar (press N if the sidebar is hidden)
- [ ] Your avatar is selected (click it in the viewport or the outliner)

Phase 1 — Fix Model

The Fix Model step is the mandatory foundation for everything else in CATS. Run it first, even if your model looks fine. It removes hidden problems that would otherwise silently break the tools that come after.

What Fix Model does:

- Removes duplicate vertices
- Removes loose geometry (disconnected triangles floating away from the body)
- Recalculates surface normals (the directions that determine which side of the mesh faces outward)
- Cleans up degenerate faces (triangles collapsed to a line or a point)
- Applies any un-applied scale or rotation that would confuse later tools

Steps:

1. In the CATS tab, find the Fix Model button at the top of the panel
2. Make sure your avatar's mesh is selected in the viewport
3. Click Fix Model
4. Wait for the operation to complete — for large meshes this may take a few seconds
5. Check that the Ledger now shows a checkmark () next to Fix Model

What you should be seeing: Your avatar looks identical or very slightly cleaner. The Ledger's first slot now shows . Previously grayed-out tools in the CATS panel are now available.

> If your model disappears or looks inside-out after Fix Model: Your mesh's normals were inverted. In Blender, select the mesh, enter Edit Mode (Tab), select all faces (A), then go to Mesh > Normals > Flip to correct it. Then re-run Fix Model.

Phase 2 — Viseme Generation

Requires: Fix Model

The Viseme Generator creates all 15 VRChat lip-sync mouth shapes from three base shapes you already have. If your avatar already has complete viseme shape keys, you can skip this phase — but you should still check the Viseme Mapper in the VRChat tab to confirm the keys are correctly named.

What the Viseme Generator does:

VRChat needs 15 specific mouth shapes (called visemes) to drive your avatar's lips when you speak. Most avatars only have a few basic mouth shapes. The CATS Viseme Generator mathematically combines your existing base shapes to produce the remaining 12.

The three base shapes it works from:

- A — Wide open mouth (the "ahh" shape)
- O — Rounded mouth (the "ohh" shape)
- CH — Narrow open mouth with teeth showing (the "ch" / "sh" shape)

Steps:

1. In the CATS tab, find the Viseme Generator section
2. Use the dropdowns to select which of your avatar's existing shape keys corresponds to A, O, and CH. If your keys are named differently (like `mouth_open`, `vrc.v_oh`, `mouth_wide`), select the closest match
3. Click Generate Visemes
4. CATS creates 15 new shape keys named to VRChat's standard (`vrc.v_aa`, `vrc.v_oh`, `vrc.v_ch`, etc.)
5. Check that the Ledger now shows a checkmark () next to Visemes

What you should be seeing: Your mesh now has 15 new shape keys in the Shape Keys panel (Properties Object Data Properties Shape Keys). The Ledger's second slot shows .

> If your avatar has no mouth shape keys at all: You will need to create at least A, O, and CH manually in Blender (using Edit Mode sculpting or shape key editing) before CATS can generate the rest. See the Glossary entry for Shape Key for a basic introduction.

Phase 3 — Eye Tracking

Requires: Fix Model

The Eye Tracking Setup tool configures your avatar's eye bones to work with VRChat's built-in eye tracking system. This makes your avatar's eyes move naturally and look at other players.

What Eye Tracking Setup does:

- Locates your avatar's left and right eye bones
- Renames them to VRChat's required names (`LeftEye` and `RightEye`)
- Creates the rotation constraints VRChat needs to drive eye movement
- Limits eye rotation to a natural range (prevents eyes from spinning 360°)
- Verifies both bones are positioned correctly relative to the head bone

Without the Fix Model step completed first, the eye bone detection may lock onto orphaned duplicate vertices from the original mesh rather than the live eye geometry, placing constraints at wrong positions. Users who skipped Fix Model before running Eye Tracking Setup report their avatar arriving in VRChat with both eyes locked in a downward stare that cannot be corrected without re-running the full pipeline.

Steps:

1. In the CATS tab, find the Eye Tracking Setup section
2. Click Auto-Detect Eye Bones — CATS searches your skeleton for bones whose names or positions match typical eye bone patterns
3. Verify that the Left Eye Bone and Right Eye Bone fields show the correct bones. If not, use the dropdown to select them manually
4. Click Setup Eye Tracking
5. Check that the Ledger now shows a checkmark () next to Eye Tracking

What you should be seeing: Both eye bones are renamed and now have rotation constraints visible in the Bone Constraints panel. The Ledger's third slot shows .

> If CATS cannot find eye bones: Your skeleton may not have dedicated eye bones. Some avatar formats (particularly older MMD models) use shape keys for blinking instead of bones. If that is your case, skip this phase — VRChat will fall back to shape-key-based eye animation automatically if no eye bones are found.

Phase 4 — Pose to Shape Key

Requires: Fix Model

The Pose to Shape Key tool converts your avatar's current posed position into a shape key (blend shape). This is useful for capturing custom expressions or rest poses that you want to use in VRChat's expressions menu.

Without the Fix Model step, the vertex order can contain gaps from removed duplicate vertices that have not yet been reconciled, causing the shape key to capture distorted geometry rather than the actual posed shape. Users who reached this step without Fix Model report shape keys that make the mesh explode outward when triggered in VRChat.

Steps:

1. Pose your avatar using Blender's Pose Mode (select the armature, press Ctrl+Tab, choose Pose Mode)
2. Move the bones into the expression or position you want to capture
3. Return to Object Mode (press Ctrl+Tab again)
4. In the CATS tab, find the Shape Key Tools section
5. Click Pose to Shape Key
6. Name the new shape key when prompted
7. Check that the Ledger now shows a checkmark () next to Pose to Shape

What you should be seeing: A new shape key appears in your mesh's shape key list. Set its value to 1.0 in the Shape Keys panel to verify it shows the correct pose.

> Shape Key to Basis: The companion tool, Shape Key to Basis, does the opposite — it bakes a shape key back into your mesh's neutral rest shape. Use this when you want to lock in a corrected rest pose permanently. This also requires Fix Model first; applying a shape key to a mesh with lingering duplicate vertices can merge geometry incorrectly.

Phase 5 — Apply Transforms

Requires: Fix Model , Visemes , Eye Tracking , Pose to Shape

This is the final phase. Apply Transforms freezes all pending position, rotation, and scale data into your mesh and skeleton so that everything reads as clean zero values (position 0,0,0 / rotation 0°,0°,0° / scale 1.0,1.0,1.0). VRChat's SDK requires clean transforms — un-applied scale in particular causes avatars to appear at the wrong size or to have physics that behaves incorrectly.

Applying transforms on a mesh that still has unfixed geometry (missing Fix Model), unresolved viseme shape keys, or unconfigured eye constraints will permanently bake those broken states into the mesh. Users who applied transforms before completing the pipeline report avatars that appear correctly sized in Blender but spawn at a fraction of normal height in VRChat, with no way to fix it without re-importing from source.

Transform tools in this phase:

- Apply All Transforms — Applies position, rotation, and scale to both the mesh and the armature simultaneously
- Fix FBT — Applies a specific transform correction for Full Body Tracking setups (moves the root bone to floor level)
- Remove FBT — Removes the FBT correction if you applied it by mistake or no longer need it

Steps:

1. Confirm all four earlier Ledger checkmarks are showing ()
2. In the CATS tab, find the Transform Tools section
3. Click Apply All Transforms

4 Check that the Ledger now shows a checkmark () next to Apply Transforms — all five slots should now be checked ()
5 Go to the VRChat tab Export section and export your avatar as FBX

What you should be seeing: All five Ledger slots show . Your avatar is ready to import into the VRChat Creator Companion.

What This Unlocks: A fully pipeline-processed avatar with clean geometry, all 15 visemes, configured eye tracking, any custom expressions you created, and clean transforms — the complete set of requirements for a VRChat upload that works correctly from the first attempt.

CATS Plugin — Before You Begin: The Pipeline Order

Read this before using any CATS tool for the first time.

The CATS tools are not a menu of independent options — they are a pipeline. Each step builds on the previous one. Running them out of order produces broken results that are invisible until you are already in VRChat, at which point they cannot be fixed without starting over.

The Five Phases, In Order

The CATS Ledger — visible at the top of the CATS panel — tracks your progress through these phases with checkmarks:

1 Fix Model — Clean the mesh. Remove duplicates, broken geometry, and bad normals. This is the foundation every other step depends on.

1 Visemes — Generate all 15 VRChat lip-sync mouth shapes from your three base shapes (A, O, CH).

1 Eye Tracking — Set up VRChat's eye movement system using your avatar's eye bones.

1 Pose to Shape — Capture any custom poses or expressions as shape keys.

1 Apply Transforms — Freeze all position/rotation/scale data so your avatar has clean transforms for upload.

The Ledger

The Ledger is the row of checkmarks at the top of the CATS panel. Each phase has one slot. When a phase completes successfully, its slot fills with a .

Tools that depend on an earlier phase being complete will appear grayed out (unavailable) until that phase's checkmark is shown. This prevents you from accidentally running steps out of order.

The Ledger resets when you start working on a new avatar. It is stored per-session in Blender's scene data.

Why This Order

Each phase modifies the mesh or skeleton in ways the next phase depends on:

- Fix Model must be first because it changes vertex count, removes duplicates, and corrects normals. Any tool that reads vertex positions (viseme generation, eye bone detection, pose capture) will produce wrong results if it runs against a mesh that still has duplicate or broken vertices.
- Visemes before Eye Tracking because both tools modify shape key and bone data. Running them in this order ensures shape key slots are allocated before constraint data is written.
- Eye Tracking before Pose to Shape because Pose to Shape captures the full mesh state including bone-driven deformations. Having eye constraints in place before capturing ensures the neutral eye position is correct in the saved shape.
- Apply Transforms last because it permanently freezes all data. Any unfixed geometry, unmapped shape keys, or misconfigured constraints get locked in permanently. Once transforms are applied, you cannot go back to fix earlier phases without re-importing from source.

What Breaks If You Skip a Step

Skip Fix Model: Every subsequent tool is running against a mesh that may contain duplicate vertices at the same position. The Viseme Generator will produce shape keys that control both the real vertex and its hidden duplicate — in VRChat, the duplicate vertex stays behind while the real one moves, causing a torn or glitched mouth at every lip-sync shape. Eye Tracking Setup may attach constraints to the duplicate eye mesh rather than the visible one, locking the eyes into a fixed stare. Apply Transforms will permanently bake all of these errors into the mesh with no way to recover.

Skip Visemes: The five pipeline phases are ordered, but the Ledger treats a missing Viseme checkmark as an incomplete pipeline. Apply Transforms will not run until all earlier phases are checked — this protects you from uploading an avatar with no lip sync. If you intentionally do not want CATS visemes (because you have existing ones), mark the phase complete manually using the Mark Complete button next to the Viseme Generator.

Skip Eye Tracking: Your avatar will have no eye movement in VRChat and will stare straight forward at all times. This is acceptable if your avatar has no eye bones — use Mark Complete to bypass this phase.

Skip Pose to Shape: If you have no custom expressions to save, this phase is optional. Use Mark Complete to advance to Apply Transforms without it.

Feature Reference

This section covers every tool in BoneForge in detail. Use this when you want to understand a specific feature more deeply, or when the Quick Guides do not cover your exact situation.

Each feature entry includes:

- What it does in plain language
- When you would use it
- What all the settings do

Rig UI Tools (Phase 1)

Stability: Stable | Introduced: 5.0

These tools help you manage the visual side of working with armatures — which bones are visible, how they are organized, and quick access shortcuts.

Bone Collection Panel

What it does: Shows all of your skeleton's bone groups as labeled buttons. Click a button to show or hide that group.

> A bone collection is a named group of bones. For example, you might have collections called "IK Controls," "Face Bones," and "Deform Bones." Hiding a collection makes those bones invisible in the viewport — useful for focusing on one part of the rig.

Key controls:

- Toggle button — Shows/hides the collection
- Solo button (eye icon) — Hides all other collections, showing only this one
- Show All / Hide All — Quick buttons to show or hide everything at once
- Select Bones — Selects all bones in the collection

- Reorder arrows — Move collections up and down in the list
- Rename — Give a collection a custom display name
- Icon / Color — Assign a custom icon and color to the button for visual organization
- Sections — Group multiple collections under a collapsible header

Where to find it: Review tab Collections section

Visibility Bookmarks

What it does: Saves a snapshot of which bone collections are currently visible, so you can switch between saved views instantly.

Example use: You have set up a view showing only the face bones for expression work. Save it as "Face Only." Then show everything for weight painting. Save it as "Full Rig." Now you can switch between these views with one click instead of toggling each collection manually.

Key controls:

- Save Bookmark — Saves the current visibility state with a name
- Restore Bookmark — Applies a saved state
- Color indicators — Color-coded markers next to each bookmark for quick visual identification
- Expand — Shows additional bookmark slots beyond the default four

Default bookmark buttons: FK Arms, IK Body, Face Only, Full Rig

Where to find it: Review tab Bookmarks section

Hotkey Quick Panel

What it does: Opens a floating version of the bone collection and bookmarks panel wherever your cursor is, without navigating to the sidebar.

How to use: Press Ctrl+Shift+R in the 3D viewport. The panel appears at your cursor. Click outside it to dismiss.

Where to change the hotkey: BoneForge Preferences (Edit > Preferences > Add-ons > BoneForge)

Animation Tools (Phase 2)

Stability: Stable | Introduced: 5.0

Pose Library

What it does: Stores named poses with thumbnail previews that you can apply to your avatar with one click.

Key controls:

- Save Pose — Stores the current bone positions as a named pose entry with an auto-captured thumbnail
- Apply Pose — Snaps bones to the saved pose
- Apply Blended (0–100%) — Applies the pose at partial strength, mixing with the current position
- Apply Mirrored — Applies the pose flipped left-to-right
- Delete — Removes a pose entry
- Rename — Changes a pose's display name
- Set Category — Tags the pose for filtering
- Filter — Shows only poses matching a category tag
- Refresh Thumbnail — Re-renders the preview image from the current viewport
- Export — Saves poses to a `.bfpose` file
- Import — Loads poses from a `.bfpose` file

Where to find it: Review tab Pose Library section

Rigify Enhancement

What it does: Automatically detects Rigify-generated control rigs and sets up BoneForge's collection panels, bookmarks, and property sliders to match Rigify's standard controls.

> Rigify is a Blender built-in system for generating animation-ready rigs. If you used Rigify to build your rig, this tool wires BoneForge's UI to Rigify's IK/FK controls automatically.

Key controls:

- Enable Rigify — Manually triggers enhancement on the active armature
- Auto-Enhance — Runs automatically when a Rigify rig is selected (optional toggle)
- Re-Enhance — Rebuilds the BoneForge panels from scratch for the current Rigify rig
- IK/FK slider — Blends between IK (position-based) and FK (rotation-based) control on arms and legs
- Stretch toggles — Enables or disables stretchy IK on limbs
- Parent space switches — Changes which space a limb's IK target is parented to (World, Root, etc.)
- Head/Neck follow — Controls how much the head/neck follows the body rotation

Where to find it: Setup Rigging tab Rigify section

Corrective Shape Keys

What it does: Creates shape keys (blend shapes) that automatically activate when a bone reaches a specific angle. Used to fix mesh pinching or collapsing at extreme poses.

Example use: Your character's elbow mesh collapses when fully bent. You sculpt a corrected version of that elbow and link it to the arm bone so it automatically applies at 150° of bend.

Key controls:

- Create Corrective — Dialog to set which bone drives the shape key, at what rotation angle it activates, and how smoothly it fades in (falloff range)
- Edit — Adjust the activation angle and falloff for an existing corrective
- Delete — Removes the corrective and its driver
- Rotation axis — Which axis (X, Y, or Z) triggers the shape key
- Activation angle — The rotation angle (in degrees) at which the shape key reaches full strength (1.0)
- Falloff — How many degrees before the activation angle the shape key starts fading in (larger = smoother transition)

Where to find it: Skin tab Correctives section

Graph Tools and Breakdowner

What it does: A set of animation refinement tools for working with keyframes and pose transitions.

Key tools:

- Breakdowner — Hold down the operator key and drag your mouse left-right to blend the current frame's pose between the nearest keyframes. Like an interactive "in-between" creator.
- Delta Move — Nudge selected bones by a precise amount in screen space or world space. Useful for fine-positioning during animation.
- Buffer Curves — Save the current animation curves to memory (Capture), then swap back and forth between the saved version and the edited version (Swap). Like undo/redo for animation curves only.
- Smart Bake — Bakes simulation or constraint-driven animation to keyframes with reduced keyframe density (removes redundant keys automatically)
- Euler Filter — Fixes rotation flipping artifacts in animation curves caused by gimbal lock
- Tangent Tools — Set keyframe handle types (Auto, Vector, Aligned, Free) on selected keyframes

Where to find it: Review tab Graph Tools section

Weight Tools (Phase 2B)

Stability: Stable | Introduced: 5.0

These tools control how your mesh deforms when bones move. Think of weights as the instructions telling each part of your mesh which bones to follow — and by how much.

Weight Mirror

What it does: Copies weights from one side of your avatar to the mirror-opposite side. Essential for symmetrical characters.

Key controls:

- Mirror All Weights — Mirrors every bone group to its opposite side
- Mirror Active Weight — Only mirrors the currently selected bone group
- Axis — Which axis is the mirror plane (X is standard for humanoids facing +Y)
- Direction — Bidirectional (copy both ways), Left to Right (left side is the source), Right to Left
- Search Distance — Maximum distance (in Blender units) to consider two vertices a "pair." Increase if your mesh is not perfectly symmetrical
- Normalize After — Ensures all weights sum to 1.0 after mirroring

Where to find it: Skin tab Weight Mirror section

Weight Transfer

What it does: Copies weights from one source mesh or bone group to a target. Used when attaching clothing to a body rig, or copying weights from a high-resolution mesh to a lower-resolution one.

Key controls:

- Source Group — The bone group to copy FROM
- Target Group — The bone group to copy TO
- Threshold — Minimum weight value to transfer (lower values = transfer more, including faint influence)
- Normalize After Transfer — Keeps all weights summing to 1.0

Transfer method:

- Nearest Vertex — Each target vertex gets the weight from the closest source vertex
- Nearest Face — Uses face projection for smoother results on curved surfaces

Where to find it: Skin tab Weight Transfer section

Weight Table

What it does: A spreadsheet-style view showing exact weight values for each selected vertex against each bone. Lets you type in precise numbers.

How to use: Select vertices in Edit Mode, then open the Weight Table. Each row is a vertex, each column is a bone. Click any cell and type a new value (0.0 to 1.0).

Key controls:

- Set Weight — Apply a typed value to a specific vertex/bone cell
- Zero Weight — Clear a specific cell to 0.0
- Tag Deform Bones — Marks selected bones as deform bones (needed for them to appear in weight paint mode)

Where to find it: Skin tab Weight Table section

Delta Mush

What it does: Applies a smoothing deformation to your mesh that reduces pinching and collapsing at joints. The mesh stays close to its original shape at rest, but deforms more cleanly during movement.

Key controls:

- Add Delta Mush — Adds the Delta Mush modifier to your mesh
- Bind — Bakes the current rest shape, anchoring the smoothing to that baseline
- Remove — Removes the modifier
- Iterations — How many smoothing passes to apply (higher = smoother, but may lose detail)
- Influence — How strong the smoothing effect is (0 = off, 1 = full strength)

Where to find it: Skin tab Delta Mush section

Proximity Wrap

What it does: Makes one mesh follow the surface of another mesh closely, like a second skin. Useful for clothing that needs to hug a body tightly.

Key controls:

- Bind — Attaches the clothing mesh to the body mesh using proximity detection
- Rebind — Re-calculates the attachment with different settings
- Unbind — Removes the proximity wrap link
- Target Mesh — The mesh the clothing should follow
- Max Distance — How far from the target surface the wrapping effect reaches
- Falloff — How the wrapping effect fades off at the edges (Smooth or Linear)

Where to find it: Skin tab Proximity Wrap section

Shape Library

What it does: Stores and retrieves shape key states (blend shape configurations). Save a set of active shape keys as a named preset and reapply it later.

Key controls:

- Save Shape — Records the current shape key values as a named entry
- Apply Shape — Sets your mesh's shape keys to match a saved entry
- Copy Shape From — Copies a shape key from another object into your current mesh

Where to find it: Skin tab Shape Library section

Rig Controls (Phase 2C)

Stability: Mixed — see individual entries | Introduced: 5.5+

Space Switching

What it does: Lets you change which "space" a bone is anchored to during animation. For example, a hand holding a prop can be switched from following the body (body space) to staying in place in the world (world space) with one click.

Stability: Stable

Key controls:

- Add Space — Creates a new space option for the active bone (name it, set type to World/Origin/Bone, set which bone to follow)
- Remove Space — Deletes a space option
- Switch Space — Moves the bone to the selected space and adds a keyframe
- Switch Without Key — Switches space without keyframing
- Set Default Space — Sets which space the bone starts in

Where to find it: Review tab Space Switching section

Spline IK

What it does: Creates a Spline IK setup — a system where a chain of bones follows the shape of a curve. Used for tails, tentacles, ropes, long hair strands, or spines that need smooth, sweeping movement.

Stability: Stable (bug-fixed in 6.1.1)

Key controls:

- Generate Spline IK — Creates the curve and IK constraint on your selected bone chain
- Remove Spline IK — Removes the setup
- Start/End Bone — First and last bones of the chain
- Curve Resolution — How many segments the control curve has (more segments = smoother but heavier)

Where to find it: Review tab Spline IK section

Chain Dynamics

What it does: Applies physics-like secondary motion to a bone chain. The bones simulate inertia — they lag behind when the parent moves and bounce back when it stops. Used for hair strands, tails, and accessories.

Stability: Stable

Key controls:

- Add Chain Dynamics — Attaches dynamics to a bone chain
- Remove Chain Dynamics — Removes them
- Bake Chain Dynamics — Converts the simulated motion into keyframes (required for export)
- Stiffness — How resistant the chain is to bending
- Damping — How quickly the motion settles
- Gravity — Downward pull on the chain

Where to find it: Review tab Chain Dynamics section

Ribbon / Bendy Bones

What it does: Creates a ribbon-style deformation system using Bendy Bones — a Blender feature that lets a single bone segment curve and twist smoothly. Good for lips, eyebrows, belts, and other soft curved areas.

Stability: Stable

Key controls:

- Generate Ribbon — Creates the ribbon bone structure
- Remove Ribbon — Removes it
- Segment Count — Number of sub-divisions along the ribbon
- Twist Amount — How much the ribbon can twist end-to-end

Where to find it: Review tab Ribbon section

Viseme / Lip Sync System

What it does: Creates and manages viseme (mouth shape) sets linked to shape keys. For VRChat viseme mapping, use the VRChat Viseme Mapper instead. For generating visemes from scratch, use the CATS Viseme Generator.

Stability: Stable

Key controls:

- New Viseme Set — Creates a named collection of viseme entries
- Record Viseme — Saves the current shape key state as a viseme
- Preview Viseme — Plays back a viseme's shape key values
- Delete Set — Removes a viseme set

Where to find it: Review tab Viseme section

SDK / Custom Drivers

What it does: Creates links between bone positions and shape keys without using Python expressions. Move a bone to a specific position, record that as a keyframe, and assign a shape key value — BoneForge creates the driver curve automatically.

Stability: Experimental

Example use: Move the eyebrow bone up by 10 units shape key "Brow Raised" = 1.0. Move it back to rest shape key = 0.0. Now the shape key follows the bone automatically.

Key controls:

- Create Driver — Opens a dialog to set source bone, target shape key, and the axis/distance to measure
- Edit Driver — Modify an existing driver
- Delete Driver — Removes it
- Record Point — At the current bone position, records the current shape key value as a point on the driver curve
- Set Driver Value — Manually types a shape key value for a recorded point

Where to find it: Review tab SDK Author section

Rig Validator

What it does: Checks your rig against a set of rules and reports any problems — naming errors, missing bones, bad hierarchy, weight issues, and VRChat-specific requirements.

Stability: Stable

Key controls:

- Run Validation — Executes all checks and shows results
- Select Bone — Jumps to the bone that failed a specific check
- Export Report — Saves the validation results as a text or Markdown file
- Rule Set — Choose Standard (general rigging rules) or VRChat (VRChat-specific requirements)

Where to find it: Review tab Rig Validator section

Rig Notes

What it does: Lets you attach written notes to your rig file — useful for documenting what you have set up, leaving reminders, or collaborating with others.

Stability: Stable

Key controls:

- Add Note — Creates a new note with a title and text body
- Edit Note — Modifies existing text
- Remove Note — Deletes a note
- Rig Readme — Displays notes in a formatted, read-only view

Where to find it: Review tab Rig Notes section

Auto-Rigging (Phase 3)

Stability: Stable | Introduced: 6.0

Auto-Rig Wizard

What it does: A step-by-step guided process that places marker points on your mesh and automatically generates a complete skeleton with weights. The main way to create a new rig from scratch in BoneForge.

See Guide 1: Get Your First Avatar Into VRChat for a complete walkthrough.

Steps: Select Mesh Set Rig Type Set Finger Count Place Body Markers Place Face Markers Place Finger Markers Review Generate

Key wizard controls:

- Guess Markers — Auto-detects marker positions from mesh geometry
- Place Marker — Interactive point placement in the 3D viewport
- Move Marker — Reposition a placed marker
- Reset Marker — Clears a marker back to unplaced
- Mirror — Auto-mirrors markers across the centerline when placing
- Confirm All Green — Locks in all green (valid) markers at once
- Kinematics — Chooses whether the generated rig uses IK + FK, IK only, or FK only
- Generate Control Shapes — Adds viewport-friendly custom shapes for generated controls
- Spine Segments — Sets the number of generated spine bones, from 2 to 8
- Neck Segments — Sets the number of generated neck bones, from 1 to 4
- Back / Next — Navigate wizard steps
- Generate — Creates the armature from confirmed markers
- Cancel — Abandons the wizard and undoes any changes

8.3.1 IK behavior: In IK + FK and IK Only modes, BoneForge creates dedicated hand and foot IK target controls named **`hand_ik.L`**, **`hand_ik.R`**, **`foot_ik.L`**, and **`foot_ik.R`**. These controls do not deform the mesh. They are used by IK constraints so the hands and feet can be positioned cleanly without making the deforming hand or foot bones serve as their own targets.

Where to find it: Rig Builder tab Wizard section

Quick Human

What it does: Generates a complete human rig with one click using preset defaults. Faster than the Wizard but less customizable.

Key controls:

- Generate Quick Rig — Creates a default human skeleton, weights, and BoneForge panels immediately

Where to find it: Rig Builder tab Quick Rig section

Mannequin Generator

What it does: Creates a stylized human figure mesh with adjustable body proportions. Useful as a starting reference when you do not have a 3D model yet.

Stability: Stable

Key controls:

- Add Mannequin — Opens proportion settings and generates the figure
- Quick Mannequin — Generates with default proportions immediately
- Regenerate — Rebuilds with different settings

- Remove — Deletes the mannequin and its rig
- Gender — Male or Female body proportions
- Height — Total height in centimeters (120–220 cm range)
- Head proportion — Relative head size
- Torso/Arm/Leg proportions — Relative length adjustments
- Muscularity — Body type from lean to heavily built

Where to find it: Rig Builder tab Mannequin section

Animation Retargeting

What it does: Takes an animation (a series of keyframed poses) from one skeleton and applies it to a different skeleton. Lets you use Mixamo animations, motion capture data, or any other animation source on your custom rig.

Stability: Stable

Key controls:

- Select Clip — Choose an animation to retarget
- Import Clip — Load animation from a file
- Auto-Match Bones — Detects matching bones between the source and target skeletons by name
- Preview — Plays the retargeted animation in the viewport
- Apply — Writes the retargeted motion as keyframes on your rig
- Bone Mapping editor — For each source bone, specify which target bone receives its motion
- Retarget Method — Simple (direct rotation transfer) or IK-Aware (accounts for limb length differences)
- Frame Range — Start and end frame of the animation to import

Where to find it: Setup Rigging tab Retargeting section

Bone Merge

Stability: Stable | Introduced: 6.0

See Guide 11: Merge Two Rigs Together for a complete walkthrough.

The three stages:

1. Scope (Stage 1) — Analyze and review the differences between two armatures
2. Rename (Stage 2) — Resolve naming conflicts and mark unique bones
3. Execute (Stage 3) — Dry-run preview, then merge

Key controls:

- Source Armature — The secondary skeleton being absorbed
- Target Armature — The main skeleton that survives
- Analyze — Compares the two skeletons and creates the diff table
- Normalize — Auto-renames all standard bones to a consistent naming convention
- Propose — Suggests names for unrecognized source-only bones
- Apply Rename — Renames one bone entry (one undo step)
- Batch — Applies a naming pattern to multiple entries at once (supports `{bone}`, `{side}`, `{index}` tokens)
- Mark Unique — Flags a bone as intentionally new (will be added, not merged)
- Dry Run — Shows what the merge would do without changing anything
- Execute Merge — Creates a backup and performs the merge

Naming standards: Mixamo Prefixed, Mixamo Stripped, or Custom

Where to find it: Review tab Bone Merge section

VRChat Tools

Stability: Stable | Introduced: 5.0, expanded 6.0

Humanoid Mapper and Validator

Maps your skeleton's bones to VRChat's required humanoid slots and checks for errors.

See Guide 5: Map Your Avatar to VRChat's Body System.

Hair Physics

Generates PhysBone components for dynamic hair and accessories.

See Guide 6: Add Hair Physics.

Clothing Merge

Attaches clothing meshes to your base avatar's skeleton.

See Guide 7: Attach Clothing That Moves With Your Body.

Naming Conventions

What it does: Detects your skeleton's naming format and renames bones to VRChat's standard.

See Guide 4: Fix Your Avatar's Bone Names.

Available presets: Mixamo, Ready Player Me, Unity Standard, Custom (save your own)

Batch tools: Add Prefix, Remove Prefix, Add Suffix, Remove Suffix, Find & Replace (plain text and regular expression)

Where to find it: VRChat tab Naming section

Viseme Mapper

What it does: Maps your mesh's shape keys to VRChat's 15 lip-sync phonemes.

See Guide 8: Set Up Lip Sync.

For generating visemes from scratch when your avatar does not have them yet, see CATS Tool Reference: Viseme Generator.

VRChat's 15 phonemes: `aa`, `ch`, `dd`, `e`, `ff`, `ih`, `kk`, `mm`, `nn`, `oh`, `r`, `ss`, `th`, `uh`, `pp`

Performance and Optimization

What it does: Measures your avatar's VRChat performance rank and provides tools to improve it.

See Guide 9: Improve Your Avatar's Performance.

Performance tiers (best to worst): Excellent Good Medium Poor Very Poor

Tools:

- Calculate Rank — Estimates your current performance tier
- Decimate — Reduces polygon count by a percentage
- Remove Unused Shape Keys — Clears unmapped blend shapes
- Remove Unused Vertex Groups — Clears empty bone assignments
- Remove Zero-Weight Bones — Removes bones with no mesh influence
- Merge Same-Material Meshes — Combines meshes that share the same material
- Material Atlas — Bakes multiple materials into a single texture sheet

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Mesh Cleanup

What it does: Fixes common mesh problems before export.

Tools:

- Fix Model — Removes duplicate vertices, loose geometry, and calculates correct normals automatically
- Join Meshes — Combines all meshes into one while keeping material slots
- Apply Transforms — Freezes scale/rotation so they read as 1.0/0° (required by some exporters)

Where to find it: VRChat tab Cleanup section

VRChat Export

What it does: Exports your finished avatar as an FBX file formatted specifically for VRChat's SDK.

Key settings:

- Folder — Choose the export folder for the `.fbx` and optional `.bfvrc` sidecar
- Avatar — Set the exported file name
- Sidecar — Writes a `.bfvrc` metadata file next to the FBX
- Merge Meshes — Combines mesh copies during export when you want a single mesh
- Separate Clothing — Keeps clothing as separate mesh objects when enabled
- Bake Shape Keys — Applies shape keys to the export copy before writing the FBX
- Embed Textures — Packs image textures into the FBX for easier Unity material import
- Helper Meshes — Includes hidden/render-disabled/custom-shape helper meshes only when enabled; leave this off for normal avatar exports

Where to find it: VRChat tab Export section

VRM Bridge

Stability: Stable | Requires VRM add-on | Introduced: 5.5

VRM Import

What it does: Imports `.vrm` files (VRoid Studio, Virtual Cast, and other VRM-format avatars) into Blender with their materials, skeleton, and shape keys preserved.

File > Import > VRM (.vrm)

Note: Requires the VRMC-io VRM add-on to be installed. Use BoneForge's VRM installer (VRM tab Install VRM Add-on) for easy setup.

VRM Export

What it does: Exports your rigged character back to VRM format for use in VRoid-compatible apps, Virtual Cast, or Resonite.

Key settings:

- Folder — Choose where the exported VRM or target FBX is written
- File — Set the output file name; BoneForge chooses the extension from the target
- Target — Choose VRM 1.0, VRM 0.x, VRChat FBX, VSeeFace, Warudo, or Resonite
- Scope — Export the active armature, or every armature with preserved VRM metadata
- Skip Lint on Export — Bypasses target validation; use only when you understand the reported risk
- Author / License — Creator information stored in the VRM metadata

Where to find it: VRM tab Export section

VRM Linter

What it does: Validates the active armature against the selected target before export. The linter checks required humanoid mapping, VRM metadata, target-specific visemes, and host-specific expectations for VRM 1.0, VRM 0.x, VRChat FBX, VSeeFace, Warudo, and Resonite.

Click Lint Now to run the check without exporting or changing your model. Errors block export unless Skip Lint on Export is enabled; warnings explain issues that may still import but could behave worse in the target app.

If the linter says required humanoid bones are missing even though the bones exist, click Fix Humanoid Map. BoneForge auto-detects the humanoid slots, saves the mapping, and stamps **`boneforge_humanoid_alias`** on the correct bones. This fixes stale mapping data without renaming the actual bones.

Where to find it: VRM tab Lint section

MMD Bridge

Stability: Stable | Requires MMD Tools add-on | Introduced: 5.5

MMD Import

What it does: Imports MMD model files (**`.pmx`**, **`.pmd`**) into Blender with bone structure and materials.

Supported formats:

- `.pmx` / `.pmd` — MMD model files
- `.vmd` — MMD animation files
- `.vpd` — MMD pose files

Note: Requires MMD Tools to be installed. Use BoneForge's MMD installer (MMD tab Install MMD Tools) for easy setup.

MMD Export

What it does: Exports your work back to PMX/VMD/VPD format for use in MMD Studio or other MMD-compatible software.

Key settings:

- Folder — Choose the destination folder for PMX, VMD, and VPD exports
- PMX File / PMX Scope — Export the active MMD model or every MMD model in the scene
- VMD File / VMD Scope — Export active-scene motion for one or all MMD models
- VPD File / VPD Scope — Export the current pose for one or all MMD models

Where to find it: MMD tab Export section

I/O Hub (Export Hub)

Stability: Stable | Introduced: 6.0

Export Hub

What it does: A central panel for all export formats — VRChat (FBX), VRM, MMD (PMX), Unreal Engine (FBX), and Unity.

Target options:

- VRChat (Unity FBX) — Standard VRChat export with folder/name settings, sidecar output, embedded textures, and helper mesh filtering
- VRM — Delegates to the VRM exporter with target, folder, file, scope, and lint controls
- MMD (PMX/VMD/VPD) — Delegates to MMD Tools with folder, file, and scope controls for model, motion, and pose exports
- Unreal Engine FBX — FBX with Unreal-specific scale, selected-only, leaf-bone, animation, and embedded-texture settings

- Unity General — Use the VRChat/Unity FBX path for SDK import; embedded textures help Unity's material import find image data

Common FBX settings:

- Folder / File — Pick the output location and file name before pressing export
- Selected Only / Scope — Choose whether the active rig, selected objects, or all matching models are exported
- Bake Animation — Converts animation to FBX keyframes when the target needs it
- Embed Textures — Packs image textures into FBX files for easier Unity/Unreal material import

Where to find it: In the sidebar under the I/O Hub tab (registered at the bottom of the sidebar)

Bridge Manager

What it does: Checks which format-bridge add-ons (VRM, MMD) are currently installed and their versions. Shows installation buttons for any that are missing.

Where to find it: VRM tab or MMD tab top section

Taskboard

Stability: Stable | Introduced: 6.0

Project Overview Panel

What it does: Shows a summary of the current avatar project — the avatar's name, a health indicator, and a list of pending tasks detected by BoneForge's analyzer.

The task analyzer automatically identifies common issues (missing humanoid bones, unresolved naming, missing visemes, etc.) and lists them as actionable items.

Where to find it: Review tab Overview section

Bone Inspector

What it does: Shows detailed information about the currently selected bone — its name, parent, constraints, custom properties, and drivers. Also lets you edit basic properties directly without entering Edit Mode.

Key information shown:

- Bone name and parent
- Constraint list (click to expand details)
- Driver list (click to open the driver editor)
- Custom property values (editable inline)

Where to find it: Review tab Bone Inspector section

Bone Context Menu

What it does: Adds BoneForge-specific options to the right-click context menu when you right-click a bone in the viewport or outliner. Quick access to common per-bone operations without opening a panel.

Automatically available when BoneForge is installed.

CATS Tool Reference

Stability: Stable | Introduced: 7.1.1

The CATS tools live in their own CATS tab in the N-panel sidebar. They are separate from the main BoneForge tabs. All CATS tools operate within the pipeline system described in CATS Plugin — Before You Begin: The Pipeline Order.

Fix Model

Pipeline phase: Phase 1 (mandatory first step) Ledger slot: 1 of 5

What it does: Performs a comprehensive one-click mesh cleanup before any other CATS operation. Removes hidden problems that would silently corrupt every subsequent step.

Operations performed:

- Merges duplicate vertices at the same position
- Removes loose geometry (disconnected faces not attached to the main body)
- Recalculates face normals (fixes inside-out surfaces)
- Removes degenerate faces (zero-area triangles and lines)
- Removes doubles on the UV map
- Applies all pending scale and rotation transforms

Key controls:

- Fix Model — Runs all operations above on the selected mesh in one pass
- Threshold — Merge distance for duplicate vertex detection (default: 0.0001 Blender units). Increase if vertices are slightly misaligned; decrease if you want to avoid merging vertices that are close but intentionally separate

Where to find it: CATS tab Fix Model section (top of panel)

Bone Name Translation

What it does: Detects the source language of your skeleton's bone names and translates them to English VRChat-compatible names.

Supported source languages:

- Japanese () — Most common, used by MMD and many VRChat community models
- Chinese () — Simplified and Traditional
- Korean ()
- Portuguese (Português)
- Spanish (Español)
- French (Français)

The translation uses a built-in dictionary of known bone name patterns for each language. It does not require internet access and runs entirely offline.

Key controls:

- Auto-Detect Language — Analyzes the current bone names and identifies the source language automatically
- Translate Bone Names — Applies the translation after detection
- Manual Language Select — Override the auto-detected language if it chose incorrectly
- Preview — Shows a before/after comparison without committing changes

Note on scope: Bone Name Translation handles the language of the source model's bone names, not the language of your Blender interface. If you have a Japanese MMD model you want to use in the English version of VRChat, use this tool regardless of which language Blender is set to.

Where to find it: CATS tab Bone Name Translation section

Zero Weight Bone Cleanup

What it does: Finds and removes bones that have zero influence over the mesh — bones that exist in the skeleton but do not move any vertices. These bones waste performance budget without contributing anything visible.

Key controls:

- Find Zero Weight Bones — Scans the skeleton and lists all bones with zero mesh influence
- Remove Selected — Deletes the bones you have checked in the list
- Remove All Found — Removes all zero-weight bones at once
- Threshold — Minimum weight sum to consider a bone "non-zero." Default is 0.001; lower values keep more bones

When to use: After attaching clothing or merging armatures, extra bones often get carried over without being assigned to any mesh. Run this after Join Meshes and before exporting.

Where to find it: CATS tab Bone Tools section Zero Weight Bones

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Join Meshes

What it does: Combines all separate mesh objects in your scene into a single unified mesh. VRChat performs best with one mesh per avatar.

Shape key conflict handling: When joining meshes that have different sets of shape keys, CATS automatically resolves conflicts by padding missing shape keys on each mesh with a neutral (zero-value) shape, ensuring the final merged mesh has a consistent shape key set across all vertices.

Key controls:

- Join All Meshes — Merges every mesh object in the scene into one
- Join Selected — Merges only the currently selected mesh objects
- Merge by Material — Joins only meshes that share a material (useful for partial merges)

When to use: After all clothing and accessories are attached and weighted. Do not run Join Meshes before CATS Fix Model — joining meshes before cleanup can spread duplicate vertex problems from one mesh object to another, making them harder to remove. Users who joined meshes before Fix Model report the resulting single mesh retaining phantom vertices from all original objects, causing the Viseme Generator to produce shape keys that visibly tear the mesh apart at the seams.

Where to find it: CATS tab Mesh Tools section Join Meshes

Material Atlas Combiner

What it does: Bakes multiple materials into a single texture atlas sheet. Fewer materials = better VRChat performance rating.

This is the same atlas process available in the main VRChat tab, presented with an Accept/Revert workflow that lets you preview the result before committing.

Key controls:

- Analyze — Shows your current material count and estimated savings
- Atlas Resolution — Size of the combined texture output (1024 / 2048 / 4096 pixels)
- Bake Atlas — Combines all materials and shows a preview
- Accept — Commits the atlas and replaces your original materials
- Revert — Undoes the atlas and restores your original materials

Where to find it: CATS tab Material Atlas section

Eye Tracking Setup

Pipeline phase: Phase 3 Ledger slot: 3 of 5 Requires: Fix Model

This tool requires Fix Model to be completed first. Without it, the eye bone detection may latch onto remnant duplicate geometry from the head mesh instead of the actual eye bone, placing rotation constraints at a point in empty space. Users who ran Eye Tracking Setup before Fix Model describe their avatar looking permanently downward at the floor with no way to correct it in VRChat without redoing the full pipeline.

What it does: Locates your avatar's eye bones, renames them to VRChat's required names (**`LeftEye`** and **`RightEye`**), and creates the rotation constraints that drive natural eye movement in VRChat.

Key controls:

- Auto-Detect Eye Bones — Searches for bones matching common eye bone name patterns and positions
- Left Eye Bone / Right Eye Bone — Manual dropdowns to assign the correct bones if auto-detect fails
- Setup Eye Tracking — Renames bones and creates all required constraints
- Eye Rotation Limits — Maximum rotation angle for up/down and left/right movement (default: 30°)
- Test Eye Movement — Animates the eye bones through their range to verify constraints work

Where to find it: CATS tab Eye Tracking Setup section

Shape Key Tools

Pipeline phase (Pose to Shape): Phase 4 Ledger slot: 4 of 5 Requires: Fix Model

Both tools in this section require Fix Model to be completed first. Capturing a shape key from a mesh that still contains duplicate vertices records both the real vertex and its hidden duplicate — when the shape key is later triggered in VRChat, users report the mesh tearing at the face as duplicate vertices pull in opposite directions.

Pose to Shape Key

What it does: Captures the current posed position of your avatar's mesh (including all bone deformations) and saves it as a new shape key. Use this to create custom expressions, clothing morphs, or alternate rest positions.

Steps:

1. Pose your avatar in Pose Mode
2. Return to Object Mode
3. Click Pose to Shape Key
4. Name the shape key when prompted
5. Verify the result by setting the new key's value to 1.0

Key controls:

- Pose to Shape Key — Captures current deformed state as a new shape key
- Name — Name field for the new shape key

Where to find it: CATS tab Shape Key Tools section

Shape Key to Basis

What it does: Bakes an existing shape key back into the mesh's neutral rest position. Effectively applies the shape key permanently as the new default pose.

Use carefully: This is a one-way operation. The shape key is removed and its deformation becomes the new base mesh shape. Make sure to run Fix Model first — applying a shape key to a mesh with lingering duplicate vertices can cause those vertices to merge at incorrect positions permanently.

Key controls:

- Shape Key to Basis — Bakes the selected shape key into the rest mesh and removes the key

Where to find it: CATS tab Shape Key Tools section

Transform Tools

Pipeline phase (Apply Transforms): Phase 5 Ledger slot: 5 of 5 Requires: Fix Model , Visemes , Eye Tracking , Pose to Shape

Apply Transforms is the final pipeline step. Running it before all earlier phases are complete bakes the incomplete state into the mesh permanently — there is no undo that can recover earlier pipeline data once transforms are applied and the file is saved. Users who applied transforms mid-pipeline describe having to re-import their avatar from source and restart the entire process from Fix Model.

Apply All Transforms

What it does: Applies position, rotation, and scale to both the mesh and the armature simultaneously, setting all transform values to clean zero/identity (location 0,0,0 / rotation 0°,0°,0° / scale 1,1,1). Required for correct behavior in VRChat's SDK.

Key controls:

- Apply All Transforms — Applies to both mesh and armature at once

Where to find it: CATS tab Transform Tools section

Fix FBT

What it does: Applies a transform correction specifically for Full Body Tracking setups. Moves the root bone so it sits at floor level, which is required for VRChat's FBT calibration system to work correctly.

When to use: Only if you intend to use Full Body Tracking with your avatar. Run after Apply All Transforms.

Key controls:

- Fix FBT — Applies the FBT root bone correction

Where to find it: CATS tab Transform Tools section

Remove FBT

What it does: Removes the FBT correction added by Fix FBT. Use this if you applied Fix FBT by mistake or no longer want FBT support on the avatar.

Key controls:

- Remove FBT — Reverts the FBT root bone adjustment

Where to find it: CATS tab Transform Tools section

Viseme Generator (CATS)

Pipeline phase: Phase 2 Ledger slot: 2 of 5 Requires: Fix Model

This tool requires Fix Model to be completed first. Generating visemes on a mesh with duplicate vertices produces shape keys that control both the actual mouth vertices and any hidden duplicates beneath them — in VRChat, the duplicates hold their original position while the real vertices move, creating a torn or split-mouth artifact at every phoneme. Users who skipped Fix Model before running the Viseme Generator consistently report a mouth that appears to rip apart at the corners when speaking.

What it does: Mathematically generates all 15 VRChat lip-sync viseme shape keys from three base shapes you define. The generator uses weighted coefficient blending so each output viseme looks like a natural combination of the base shapes rather than a mechanical interpolation.

The 15 visemes generated: `vrc.v_aa`, `vrc.v_ch`, `vrc.v_dd`, `vrc.v_e`, `vrc.v_ff`, `vrc.v_ih`, `vrc.v_kk`, `vrc.v_mm`, `vrc.v_nn`, `vrc.v_oh`, `vrc.v_r`, `vrc.v_ss`, `vrc.v_th`, `vrc.v_uh`, `vrc.v_pp`

Base shapes needed:

- A — Wide open mouth ("ahh")
- O — Rounded mouth ("ohh")
- CH — Narrow teeth-showing mouth ("ch" / "sh")

Key controls:

- A Shape — Dropdown to select your "A" base shape key
- O Shape — Dropdown to select your "O" base shape key
- CH Shape — Dropdown to select your "CH" base shape key
- Generate Visemes — Creates all 15 output shape keys
- Preview — Cycles through the generated visemes so you can check the results before committing
- Blend Strength — Scales the coefficient multipliers up or down globally (1.0 = default; reduce if visemes look too extreme)

Where to find it: CATS tab Viseme Generator section

Bone Tools

What it does: A set of utility operations for managing bones in your skeleton.

Create Root Bone

What it does: Adds a root bone at the base of your skeleton (at the world origin, floor level) and parents all existing top-level bones to it. VRChat requires a root bone as the top of the hierarchy.

Key controls:

- Create Root Bone — Adds a bone named `Root` at position 0,0,0 and re-parents the armature hierarchy

When to use: When your skeleton has no root bone, or when the Rig Validator reports "missing root bone" errors.

Where to find it: CATS tab Bone Tools section

Merge Short Bones

What it does: Finds bones below a specified minimum length and merges them into their parent bone. Very short bones are often artifacts from import or from bone chain generation — they consume performance budget without contributing visible deformation.

Key controls:

- Min Length — Bones shorter than this value (in Blender units) are candidates for merging
- Preview — Shows which bones would be merged without committing
- Merge — Applies the merge

Where to find it: CATS tab Bone Tools section

Duplicate Bones

What it does: Creates copies of selected bones — useful for setting up twist bones, deform bone layers, or adding a copy of a control chain for a different purpose.

Key controls:

- Duplicate Selected — Creates a copy of each selected bone with a `.copy` suffix
- Mirror Duplicate — Duplicates and mirrors across the centerline, creating left/right pairs

Where to find it: CATS tab Bone Tools section

Armature Tools

Merge Armatures

What it does: Combines two separate armatures (skeletons) into one. Similar to BoneForge's Bone Merge tool but optimized for the simpler use case of merging a clothing armature into a body armature.

Key controls:

- Base Armature — The main skeleton (survives the merge)
- Merge Armature — The secondary skeleton (absorbed)

- Merge — Executes the merge
- Connect Bones — Optionally re-parents the merged bones to the base armature's nearest bone instead of keeping them as top-level bones

For complex multi-stage merges with naming conflict resolution, use BoneForge's full Bone Merge tool in the Review tab instead.

Where to find it: CATS tab Armature Tools section

Mesh Tools

What it does: Additional mesh separation utilities beyond the basic Join Meshes tool.

Separate by Materials

What it does: Splits a joined mesh into separate objects — one per material. Useful if you need to work on a specific material zone independently.

Key controls:

- Separate by Materials — Splits the active mesh by material assignment

Where to find it: CATS tab Mesh Tools section

Separate by Loose Parts

What it does: Splits a mesh at disconnected geometry boundaries — each group of connected faces becomes its own object. Useful for isolating accessories or props that were accidentally joined.

Key controls:

- Separate by Loose Parts — Splits the active mesh at geometry boundaries

Where to find it: CATS tab Mesh Tools section

Separate by Shape Keys

What it does: Splits a mesh by shape key data — separates vertices that have shape key animation from those that do not. Useful for isolating the animated face mesh from a static body when you need to work on only one.

Key controls:

- Separate by Shape Keys — Creates two objects: one with shape key data, one without

Where to find it: CATS tab Mesh Tools section

CATS Validator

What it does: Checks your avatar against the CATS pipeline requirements and reports any phases that are incomplete, out of order, or have configuration problems.

The Validator is separate from BoneForge's main Rig Validator — it focuses specifically on CATS pipeline state rather than general rigging correctness.

Key controls:

- Run CATS Validation — Checks all five pipeline phases and reports status
- Jump to Phase — Opens the relevant CATS panel section for any phase that failed
- Force Reset Ledger — Clears all Ledger checkmarks and resets the pipeline to the beginning (use if you re-imported the mesh and need to run the full pipeline again)

Validation checks performed:

- Fix Model: Was it run on the current mesh? (Detects if the mesh was modified after Fix Model ran)
- Visemes: Are all 15 VRChat phoneme shape keys present and named correctly?
- Eye Tracking: Are `LeftEye` and `RightEye` bones present with correct constraints?
- Pose to Shape: Is at least one custom shape key present (or was the phase marked complete)?
- Apply Transforms: Are all transforms at clean values (scale 1,1,1 / rotation 0,0,0)?

Where to find it: CATS tab Validator section (bottom of panel)

Quick-Fix Index

Use this section when something has gone wrong and you need to find the answer fast.

Upload fails — bones not recognized	Guide 4 + VRChat Naming
Avatar in T-pose / doesn't track movement	Guide 5 + Humanoid Mapper
Mesh deforming weirdly / skin stretching	Weight Transfer + Weight Mirror
One side of body has different weights	Weight Mirror

Hair clips through head	Guide 6, Step 6 — add colliders
Hair physics not moving	Guide 6 — check chain detection + physics preset
Clothing clips through body	Guide 7 + check BVH collision detection
Lip sync not working	Guide 8 + Viseme Mapper
Avatar is Very Poor performance	Guide 9 + Performance Optimization
VRM import fails	Guide 2, Step 1 — install VRM add-on
MMD import fails	Guide 3, Step 1 — install MMD Tools
Rig validator shows red errors	Rig Validator — run validation and follow error messages
Bones vanished / can't see any bones	Bone Collection Panel click Show All
Can't find BoneForge panels	Press N in 3D viewport, look for BoneForge tabs
Export FBX is missing bones	Check armature is selected before exporting; enable "Include Armature"
Shape keys disappeared after export	Enable "Include Shape Keys" in export settings
Export blocked by missing humanoid bones	Run Auto-Map Humanoid, then Fix Humanoid Map in the VRM Lint section
Exported FBX shows giant helper shapes or tubes	Keep Helper Meshes off unless you intentionally need control/custom-shape meshes
Unity or Unreal imports gray materials	Export with Embed Textures enabled, then use Unity's material import/extract options or Unreal's FBX material import
Weights are all on the wrong bones	Re-run Auto-Weight in the Wizard, or use Weight Transfer
Two rigs need to be one	Guide 11: Merge Two Rigs Together
Corrective shape not activating	Check bone axis and activation angle in Corrective Shape Keys
Animation looks wrong on different rig	Animation Retargeting — check bone mappings
CATS tools are grayed out / unavailable	Run Fix Model first — CATS Pipeline Order
Mouth tears apart or splits when speaking	Fix Model was skipped — re-run from Phase 1 — Guide 13, Phase 1

Avatar eyes are stuck looking down in VRChat	Eye Tracking Setup ran before Fix Model — re-run from Fix Model — CATS: Eye Tracking Setup
Shape key makes mesh explode when triggered	Pose to Shape Key ran before Fix Model — re-run from Fix Model — CATS: Shape Key Tools
Avatar spawns at wrong size in VRChat	Apply Transforms ran before completing pipeline — re-import from source, restart from Fix Model
Bone names are Japanese / Chinese / Korean	CATS: Bone Name Translation
Avatar has no lip sync and no existing shape keys	CATS: Viseme Generator — generate 15 visemes from 3 base shapes
CATS Ledger checkmarks disappeared	Mesh was modified after pipeline — run CATS Validator, then re-run affected phases
Apply Transforms button still grayed out	Not all 4 earlier Ledger phases are checked — check Validator for which phase is incomplete

Glossary

Armature — Blender's word for a skeleton. A collection of bones that can be posed and animated.

Blend Shape — See Shape Key.

Bone — A single segment of a skeleton. Bones are arranged in a hierarchy (parent child) where child bones follow parent bones.

Bone Collection — A named group of bones for organizational purposes. You can show or hide entire collections at once.

CATS — The CATS Plugin, a suite of model preparation tools added to BoneForge in version 7.1.1. CATS provides a guided pipeline for cleaning, configuring, and VRChat-readying avatars. CATS lives in its own sidebar tab separate from the main BoneForge tabs.

CATS Pipeline — The five-phase ordered workflow used by the CATS Plugin: Fix Model Visemes Eye Tracking Pose to Shape Apply Transforms. Each phase must be completed before the next is available.

Corrective Shape Key — A shape key (blend shape) that automatically activates when a bone reaches a specific angle, used to fix mesh deformation at extreme poses.

Deform Bone — A bone that is marked as a deformation bone, meaning it directly influences the mesh's shape. Not all bones need to deform the mesh; some exist only as controls.

Eye Tracking Setup — The CATS tool that configures your avatar's eye bones (**`LeftEye`**, **`RightEye`**) and creates the rotation constraints VRChat uses to drive natural eye movement. Phase 3 of the CATS Pipeline.

FBT (Full Body Tracking) — A VRChat feature that uses external hardware (such as Vive Trackers) to track your full body position including hips and feet. BoneForge's Fix FBT tool adjusts the avatar's root bone for correct FBT calibration.

FBX — A file format used to transfer 3D models, skeletons, and animations between software. The standard format for VRChat.

Fix Model — The CATS tool that performs comprehensive one-click mesh cleanup: removes duplicate vertices, loose geometry, and bad normals. Always the first step in the CATS Pipeline (Phase 1). Every other CATS tool depends on Fix Model having been run first.

FK (Forward Kinematics) — A control method where you manually rotate each bone in the chain. Rotating the shoulder bone moves the arm; you then rotate the elbow, then the wrist. Natural for broad body poses.

IK (Inverse Kinematics) — A control method where you position the endpoint (like the hand) and the software automatically calculates all the intermediate bone rotations. Natural for precise hand/foot placement.

Humanoid — VRChat's built-in avatar system that maps bones to standard body positions so all avatars use the same movement controls.

Ledger — The row of checkmarks visible at the top of the CATS panel. Tracks which of the five CATS Pipeline phases have been completed for the current avatar. A filled checkmark () means the phase is done. Tools that depend on earlier phases are grayed out until the required Ledger slots are checked.

Mark Complete — A button available next to optional CATS Pipeline phases (Eye Tracking, Pose to Shape). Clicking it marks the phase as complete in the Ledger without running the tool — used when you want to skip an optional phase intentionally.

Mesh — The 3D surface geometry that makes up your avatar's visible body.

MMD (MikuMikuDance) — A free 3D animation software popular in Japan and the anime community. Uses **`.pmx`** model files and **`.vmd`** animation files.

Morph Target — See Shape Key.

PhysBone — VRChat's component for making bones simulate physics (bounce, swing, collide). Applied to hair, tails, dangling accessories, etc.

Pipeline — An ordered sequence of operations where each step depends on the previous one being correct. The CATS Plugin uses a five-phase pipeline to ensure model preparation happens in the correct order.

PMX — The main 3D model file format used by MikuMikuDance.

Pose Mode — A Blender mode for posing and animating bones. Select the armature, then press Ctrl+Tab and choose Pose Mode, or use the dropdown in the top-left of the viewport.

Rig / Rigging — The process of building a skeleton inside a 3D model and connecting the mesh to the skeleton so it can be posed and animated.

Shape Key — A saved version of a mesh in a specific deformed position. Blend shapes can be blended together or activated at different strengths. Used for facial expressions, lip sync, and body morphs.

SDK (Software Development Kit) — In VRChat context, the VRChat Creator Companion and its Unity tools for uploading and managing avatars.

Spline IK — An IK system where a chain of bones follows the path of a curve. Used for tails, tentacles, long hair strands, and spines.

T-Pose — A reference pose where the character stands upright with arms extended horizontally to the sides. Required for rigging.

Vertex — A single point in 3D space. Meshes are made of thousands of vertices connected by edges and faces.

Vertex Group — A named selection of vertices in Blender, used to define which vertices are influenced by which bone.

Viseme — A specific mouth shape associated with a phoneme (speech sound). VRChat uses 15 visemes for lip sync.

Viseme Generator — The CATS tool that mathematically creates all 15 VRChat viseme shape keys from three base shapes (A, O, CH). Phase 2 of the CATS Pipeline. Requires Fix Model first.

VMD — MikuMikuDance animation file format.

VPD — MikuMikuDance pose file format.

VRM — An open file format for 3D humanoid avatars, used by VRoid Studio and many virtual avatar platforms.

Weight / Weight Painting — The process of assigning values (0.0 to 1.0) to each vertex specifying how strongly it is influenced by each bone. Higher weight = more influence. Weight painting is the visual tool for adjusting these values.

Wizard — BoneForge's step-by-step guided rigging tool that walks you through placing markers and automatically generating a skeleton.

Zero Weight Bone — A bone in a skeleton that has no influence over any mesh vertices. These bones take up performance budget without contributing to the avatar's appearance. The CATS Zero Weight Bone

Cleanup tool removes them automatically.

BoneForge Documentation | Version 8.5.0 For support, check the BoneForge GitHub page or community Discord.